

WASP-10b

R-2028

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_121122 · created 2026-08-08T00:56:33+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456253.701 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.145 +/- 0.016
Transit depth (Rp/Rs)^2	2.11 +/- 0.47 [%]
Orbital Inclination (inc)	89.96 +/- 0.74
Airmass coefficient 1 (a1)	1.073 +/- 0.021
Airmass coefficient 2 (a2)	-0.062 +/- 0.015
Scatter in the residuals of the lightcurve fit is	2.01 %
Best Comparison Star	None
Optimal Aperture	3.76
Optimal Annulus	11.92
Transit Duration (day)	0.0933 +/- 0.0017

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.145 ± 0.016 vs 0.1592 (deviation 0.78σ)
Duration fit vs published	0.0933 d vs 0.0946 d (deviation 0.66σ)
Mid-time O–C	0.8 min
Depth SNR	8.0
Residual scatter	2.01 %

Light curve

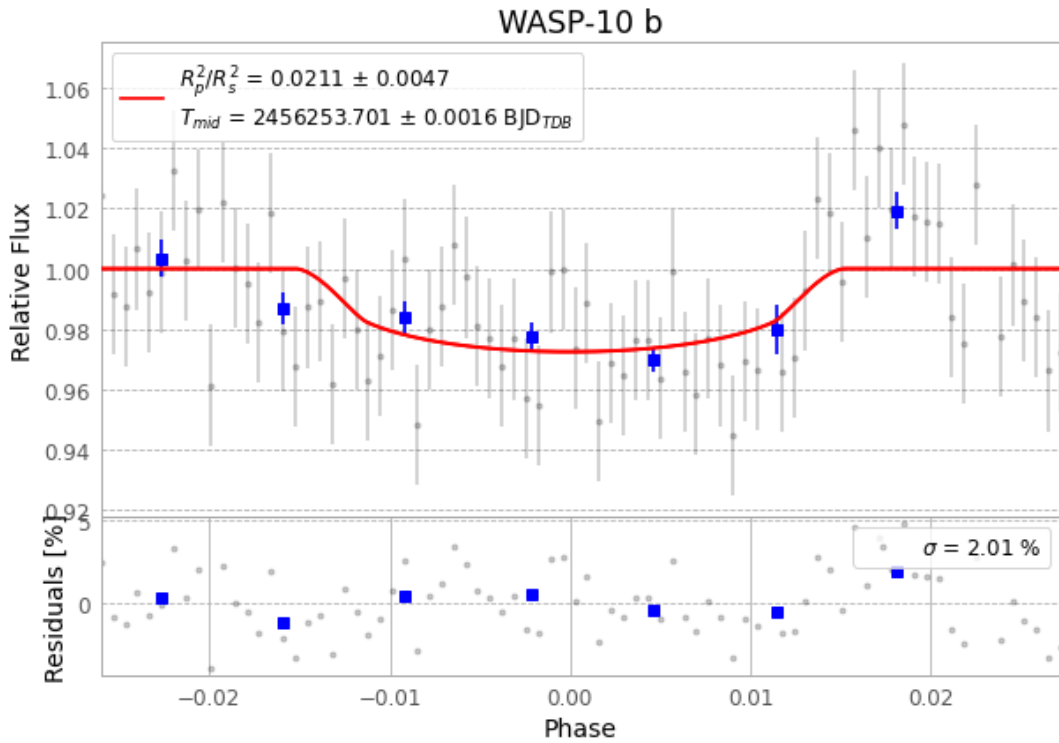


Figure 1 — FinalLightCurve_WASP-10 b_2012-11-21.png (SHA-256 b32dc82b58ff8a47...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [328, 276], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 335d36c84f6696a1...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

80 FITS frames (53 MB), WASP-10121122024812.FITS → WASP-10121122064512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-121122.log (f791042e4d9f...), FinalLightCurve_WASP-10 b_2012-11-21.png (b32dc82b58ff...), FinalLightCurve_WASP-10 b_2012-11-21.csv (ee6e7685fed5...)

Compute resources

Wall time 118.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-08 00:56Z.